

Assignment 1 Report

AI651 — Deep Learning for Space, Time and Graphs

Fall 2026

GitHub (public): <https://github.com/junaaid074/AI651-PA1>

Task 1 notebook: `Question 1/Assignment1.ipynb` (full preset, seed 0)

Task 2 code: `Question 2 - Leaderboard/src/train_leaderboard.py`

Forecast paste: `Question 2 - Leaderboard/outputs/leaderboard_predictions.txt`

Current board declarations: $P=32586$, $E=20$ (attempt 4)

1 Task 1

1.1 Response 1A

I expected period-routed ridge to win on both established stations and Station 4, then Raw Attention, then the two plain ridge models. Sensor-specific ridge should fail on Station 4 because that sensor never gets its own matrix.

Validation RMSE (m/s^2), full preset:

	Established	Station 4
Shared ridge	0.767	0.876
Sensor-specific	0.768	n/a
Period-routed	0.166	0.173
Raw Attention	0.424	0.617

That is the order I guessed, just more one-sided. Shared and sensor-specific are almost the same on established stations, so the issue is not “which sensor” — it is “what pace is this run on”. Attention can change the mix per window, so it beats a single shared map, but it still loses to a model that is given the period band. Products A/B/C tell the same story: period-routed stays near 0.15–0.18.

1.2 Response 1B

Output 1.2 is one station on a fast Product A run and a slow Product C run. One shared $\widehat{W}$ cannot follow both. Least squares splits the difference, so peaks land late or early on at least one pace.

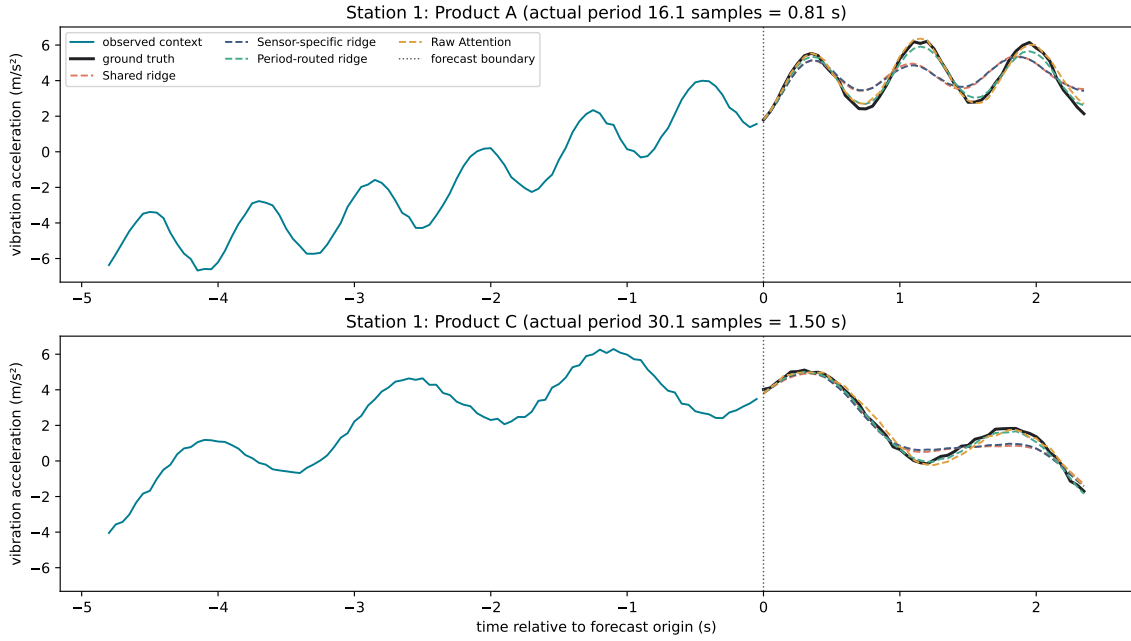


Figure 1: Output 1.2 — same station, two paces.

Output 1.3 uses the same W_Q, W_K, W_V , but the weights move when the context moves. Ridge never does that. I am not saying Attention recovered the physical period — only that the mix depends on the window.

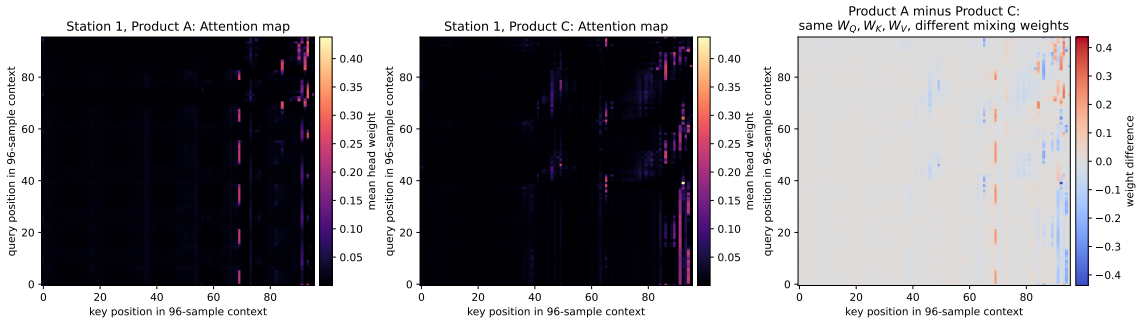


Figure 2: Output 1.3 — Attention weights change with the input.

1.3 Response 2

I implemented a centered moving average with replicate padding at the ends (zeros pull the edges down and fail the check). Return order is (remainder, trend) so $S + T = x$.

Width $m = 41$ came from Output 2.2 (best oracle period recovery on training). In Output 2.1, a wider kernel puts more of the slow ramp into T . Width 41 is still small next to the 240-sample drift, but large enough versus the 14–36 sample vibration.

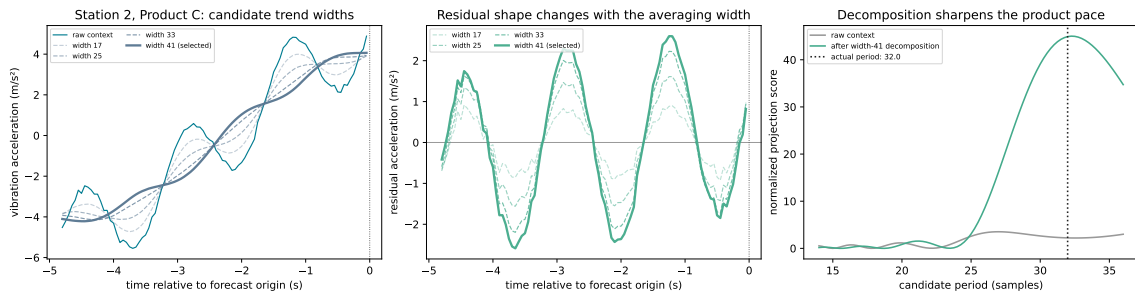
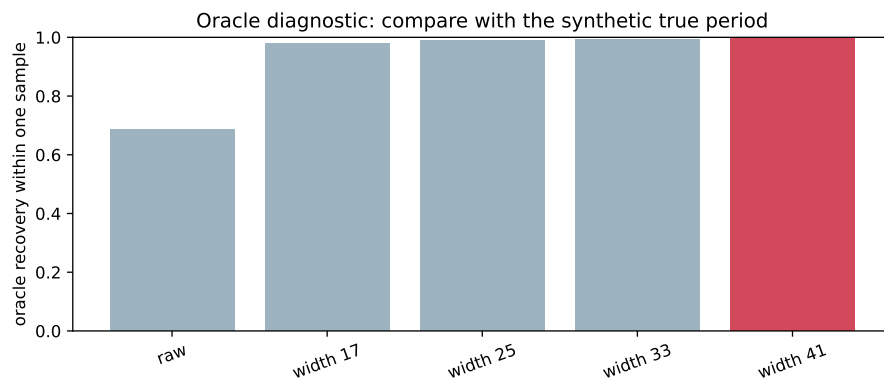


Figure 3: Output 2.1 — different widths.

Figure 4: Output 2.2 — oracle used to pick $m = 41$.

Raw Attention validation RMSE: 0.424 / 0.617 (established / held-out). With decomposition: 0.293 / 0.436. Splitting helps.

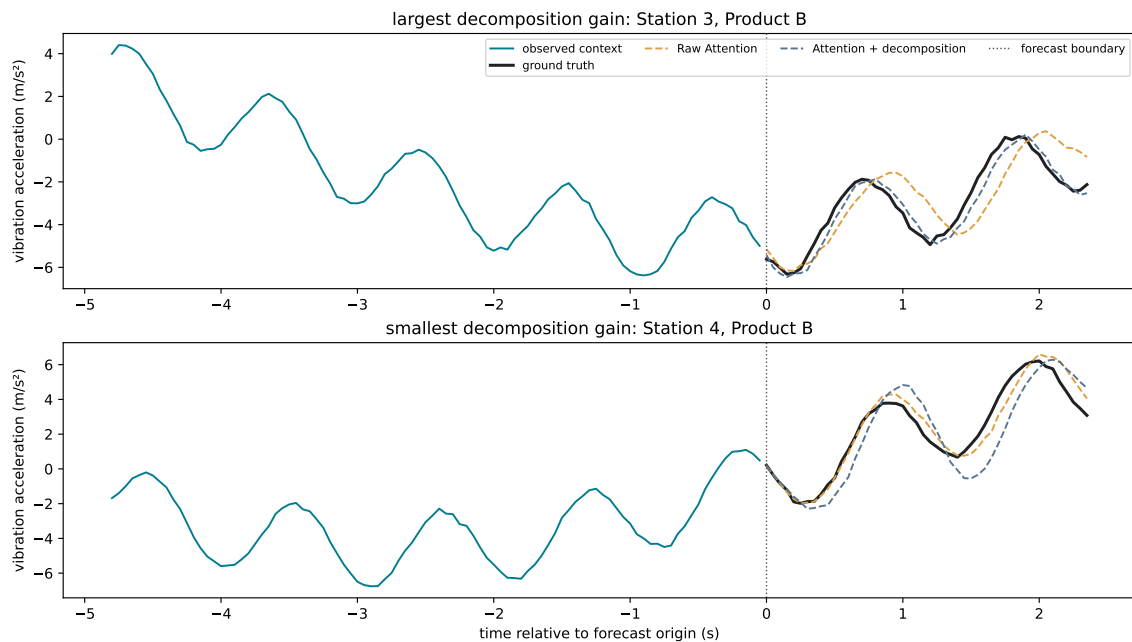


Figure 5: Output 2.3 — Attention with vs without decomposition.

Why a moving average fits this data: over one 96-sample context the slow part is smooth, and

the cycle sits on top of it. After subtracting T , the mixer mostly sees vibration. The oracle is only for choosing m ; the model never sees true $s_i(t)$.

This bias fails on a sharp jump inside the window — the jump gets smeared into the trend and then leaks into the residual. First differences would treat “trend” as local change, not local mean.

1.4 Response 3

aggregate_delays:

$$z_t = \sum_{j=1}^K \alpha_j v_{(t-\tau_j) \bmod L}.$$

I used gather so it stays differentiable. The notebook toy needs $z_0 = 35$; rolling the wrong way gives 25.

Output 3.1 matches the hand calculation. Output 3.2 is a signal check, not the trained mixer scores. After the slow part is removed, correlation peaks near P and $2P$ (delays 16 and 32 on the plotted run, residual corr ≈ 0.98). On the raw line the ramp looks similar under many shifts, so the period is harder to see. Delay mixing wants a residual stream.

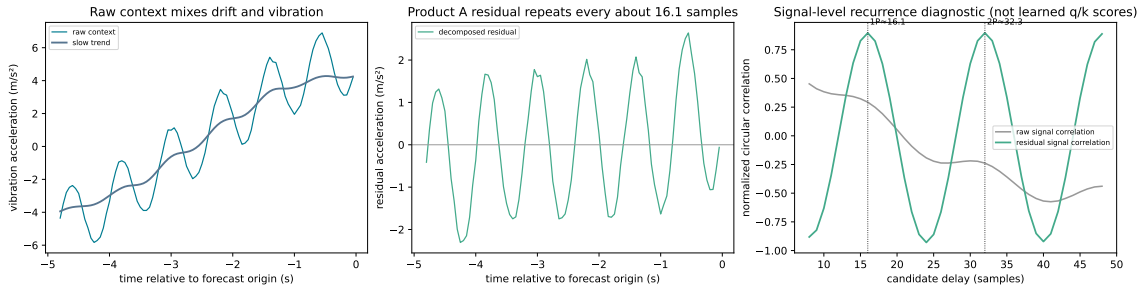


Figure 6: Output 3.2 — recurrence on raw vs residual.

Bad case: a one-off spike, or the period changing inside the same 96-sample context. Then there is no single “best delay”.

1.5 Response 4

(a) Before Output 4.1 I thought decomposition would matter most when the slow ramp is large, because otherwise Attention spends capacity on the baseline.

(b) Established validation:

Model	RMSE	MSE	#params	fit (s)
Shared ridge	0.767	0.588	4.6k	≈ 0
Sensor-specific	0.768	0.590	13.8k	≈ 0
Period-routed	0.166	0.028	59.9k	≈ 0
Raw Attention	0.424	0.180	368k	418
Attn + decomp	0.293	0.086	373k	557
Raw delay	0.226	0.051	368k	509
Autoformer-style	0.199	0.039	373k	570

Held-out RMSE: period-routed 0.173, Autoformer-style 0.244, raw delay 0.332, attn+decomp 0.436, raw attn 0.617.

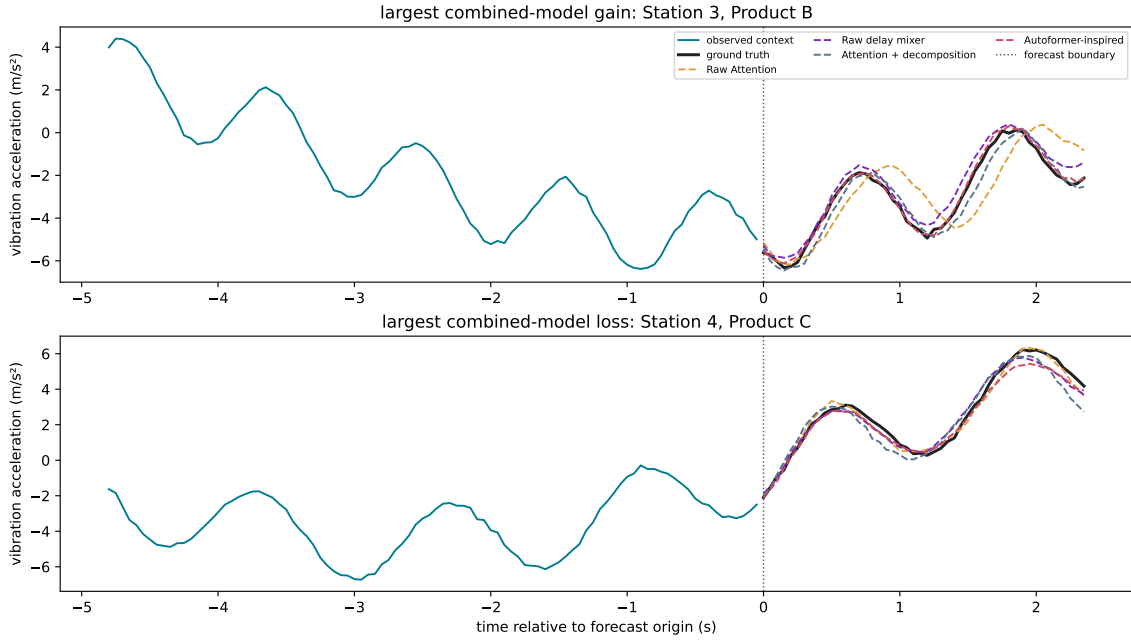


Figure 7: Output 4.1 — four neural setups.

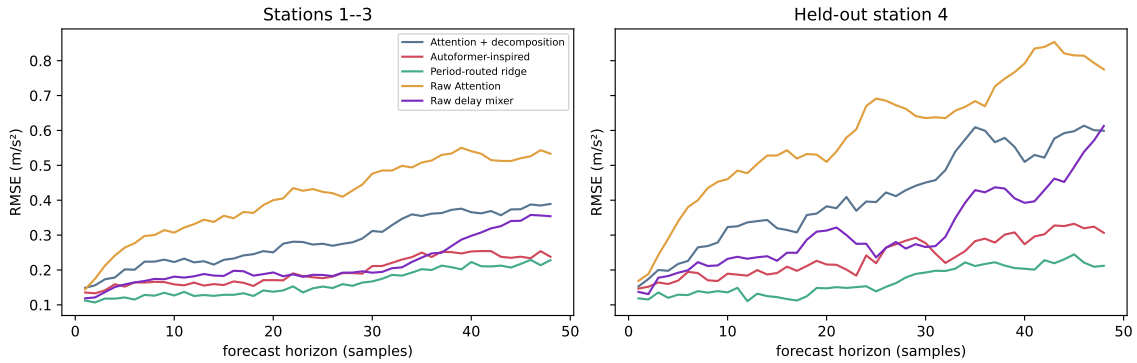


Figure 8: Output 4.2 — error vs horizon.

Both switches help. Delay mixing alone beats decomp alone. Using both is the best neural option, but the combined gain is less than adding the two separate gains — they attack related problems. Errors grow with the horizon; period routing stays lowest throughout.

(c) I deploy **period-routed ridge** for established stations and for Station 4. Best validation error, cheap to fit, and it still works on the held-out sensor. Autoformer-style is the neural winner but worse and slower. Test RMSE: 0.161 (established), 0.183 (Station 4).

Takeaway: naming the right variable (pace) beat a bigger flexible model here.

2 Task 2

The submitted model is Autoformer: moving-average decomposition plus auto-correlation / delay aggregation, `pred_len = 168`. Decomp and delay gather follow the same idea as Question 1. Extra pieces I added: $\log(1 + y)$, per-window mean/std on the target, $L = 336$, label length 84, AdamW + OneCycle, and a small linear head on the last 8 values (lag-1 is strong on this series).

I also feed the optional covariate file into both encoder and decoder (future values over the horizon are in that file). From `time_idx` I built hour-of-day and day-of-week sin/cos. There is no calendar column, but the index is hourly and those cycles showed up in simple checks.

Validation = last several non-overlapping 168-step blocks. After attempts 1–3 scored 115–120 on the board, I also held out the last training week once as a proxy, because the earlier val score was too kind.

2.1 Optional file ablation

Same small setup, chronological val RMSE:

	seed 0	seed 1
target only	106.06	108.70
+ optional file	73.20	81.02

The optional file helps on both seeds. Feature_D and feature_H were among the stronger raw correlations. The submitted run keeps that file plus the index-based hour/dow features.

2.2 Leaderboard attempts

Att.	P	E	Board RMSE	What I sent
1	32330	12	≈ 120.47	first Autoformer, covariates, no calendar
2	7530	15	—	much smaller net; local val looked good
3	7530	15	114.59	MAE 74.58, sMAPE 66.82% (best on the site so far)
4	32586	20	(this paste)	2 encoder layers + hour/dow features

Attempts 1–3 all landed near 115–120. Shrinking the net (attempt 2/3) barely moved the board. The local val I had used was leaking (refit on val blocks) and had no calendar features, so it was not a good stand-in for the hidden week.

Attempt 4 is the file in `leaderboard_predictions.txt`. Same Autoformer, $d_{\text{model}}=32$, 2 encoder layers, optional covariates, hour/dow features, 18 training epochs plus a 2-epoch refit ($E=20$). On a last-week holdout inside the training file this setup was much closer to the recent level (proxy RMSE ≈ 37 on seed 0, ≈ 41 on seed 1). That is still only a local check — the board week can differ — but it is why I am sending this one rather than another 7k-parameter run.

AI tools

I used a chat assistant for the Task 2 Autoformer wiring and for running / checking training jobs.

- **Prompt (rough):** help set up an Autoformer encoder–decoder for a 168-step forecast with series decomposition and auto-correlation, plus a training loop on the student CSV.
- **What it gave:** a skeleton model and train script.

- **What I changed:** delay gather direction after the notebook check, sinusoidal PE, the small level head, width m and the ridge deployment from my full-preset tables, and covariates / hour-dow from my own ablations.